



BUILD AN AI AGENT IN 60 MIN

From Foundation to Intelligence

GitHub Data to Intelligent Agent w/ Snowflake Cortex Code

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Tonight

- 0–5 min | Why context beats compute
- 5–10 min | CoCo orientation + environment check
- 10–30 min | Build the agent (CoCo + Cortex AI)
- 30–45 min | Make it smarter (COMPLETE + SEARCH)
- 45–55 min | Run it in CoWork
- 55–60 min | Q&A + take-home



What You'll Walk Away With

By the end of this session you will have built something real.

A Working AI Agent

- Not a slide deck. Not a demo.
- Your own CoCo-built agent, live in your Snowflake account.
- It can run against 4B+ real GitHub events.
- It answers questions in natural language.
- You own it. Take it home.

The Mental Model

- Your data → the context
- **CoCo** → the builder
- **Cortex AI** → the intelligence
- **CoWork** → the interface

This stack works on any dataset.

GitHub is just how we learn it tonight.



THE PROBLEM



General AI Tools Weren't Built for Data Teams

Your data is the moat. Tonight we build the agent that reaches it.

They're coding-only.

Cursor, Claude Code, and Copilot complete code.

They have no concept of schemas, roles, warehouses, or governance policies.

They treat Snowflake like any other database.

Their context is repo-only — and goes stale.

They read files you give them. The moment your schema changes, their context is wrong.

They can't query your live environment.

Data leaves your secure perimeter.

Pasting query results into Claude Code or Cursor means sensitive data exits your governance boundary.

With CoCo, everything stays inside Snowflake's RBAC framework.



You Already Have a Local Agent. Now Give It a Brain.

Foundation: your data in Snowflake. **Intelligence:** your agent in CoWork. **Interoperability:** runs inside the tools you already use.

Cursor / Claude Code / Copilot

- World-class **general** tools.
- Cursor: great for **app** code.
- Claude Code: great **CLI** agent.
- Copilot: great for completions.
- Shared limitation: zero
- Snowflake awareness.
- Must be manually fed context.
- Goes stale immediately.

CoCo + Snowflake (tonight)

- Works inside Cursor / VS Code.
- CLI agent like Claude Code: but Snowflake-native.
- Live context > queries your real schema right now.
- Knows your tables, roles, governance policies, compute.
- Build to CoWork to MCP server.
- Connects to any AI client.
- Tonight: 4B+ GitHub events.



Built for Data > Outperforms Claude Code on ADE-Bench

Source: ADE-Bench (Analytics and Data Engineering Benchmark by dbt), April 2026

854

Tool Calls (CoCo)

- vs 1,657 for Claude Code
- ADE-Bench: 43 tasks Analytics & Data Eng Benchmark by dbt

19.9

Avg Steps (CoCo)

- vs 38.5 for Claude Code
- Fewer steps = less wasted compute and faster iteration

15.7%

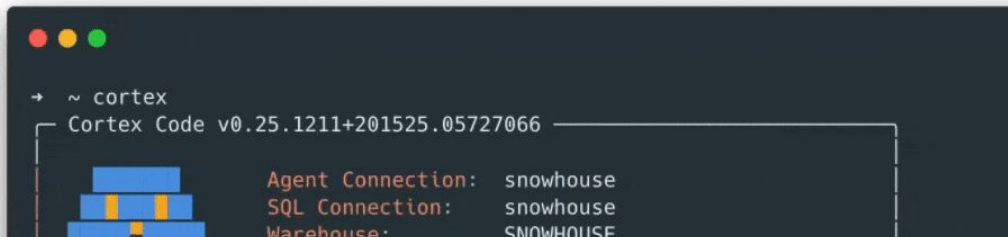
More Efficient

- Tested on Opus 4.6
- Snowflake-native context is the differentiator, not model quality

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Extra Cost

Included w/ Snowflake
vs
\$20–40/user/month
for Cursor or Claude Code



THE STACK



Your Agent Stack Tonight

3 layers. Built in 60 minutes. Yours to keep.

DATA

- GH Archive public S3 bucket
- 110M real GitHub events
Stars, forks, commits,
pull requests, issues
- Free. 1-click mount.
- Already loaded.

BRAIN

- Snowflake CoCo + Cortex AI Functions
- CoCo writes your SQL.
- CORTEX.COMPLETE turns results into language.
- CORTEX.SEARCH finds semantic matches.

INTERFACE

- CoWork or Streamlit
- Ask questions in plain English.
Get answers backed by real data.
- No SQL required.
- By anyone on your team.



LET'S BUILD



<https://github.com/sfc-gh-rbachala/building-ai-agents-with-coco-workshop>



Step 1: Load the Data

Action: Run setup SQL → COPY INTO loads ~107M events [~4 min]

Open a SQL Worksheet in Snowsight

Run the setup block from the GitHub repo >> COPY INTO loads 107M GitHub events from public S3
[~4 min.live]

What you get

EVENTS table > every public GitHub action: push, star, fork, PR, issue, comment
4B+ rows. Updated continuously. Partitioned by date.
Real signal. Real engineers. Real projects.

Why this dataset

Every person in this room has a GitHub account.
This data is about you, your repos, your community.



Step 2: Let CoCo Explore the Schema

CoCo is your pair-programming partner — fast, Snowflake-native, great at execution.
You own the design.

Open CoCo in Snowflake

Lower-right corner → Cortex Code icon.

The panel opens on the right. It already knows your account, databases, schemas.

Your first prompt

"Explore the GH Archive database. What tables exist, what columns matter, and what would be useful for finding trending AI repositories?"

4 ways to talk to CoCo

- `#GH_ARCHIVE.PUBLIC.EVENTS` — reference a specific table directly
- `/sql` — run a SQL query instantly
- `$skill-name` — invoke a bundled skill (e.g. `$cortex-agent`)
- `@path/to/file` — include a local file as context



Step 3: Build the Trending Repos Query

Bonus: Is OpenClaw in the results? Check. (It should be.)

Your CoCo prompt

"Write a query to find the top 20 most-starred repositories tagged with AI, ML, or LLM topics in the last 30 days. Include repo name, total stars gained, primary language, and description."

What CoCo generates

A working SQL query against EVENTS table

Filters for WatchEvent (star events) in the correct date range

Aggregates star counts by repo

Joins to repo metadata for description and language

Run it. See real results.

You now have a table of the actual hottest AI repos right now — not curated, not a blog post.

Real signal from 4B events.



Step 4: Add CORTEX.COMPLETE

CoCo writes the **COMPLETE** wrapper. You just describe the output you want.

Without intelligence

- You get a table.

```
repo_name: awesome-llm-agents
```

```
stars_gained: 4821
```

```
language: Python description: A curated list...
```

- Useful. But not answerable.
- A human still has to interpret it.

With CORTEX.COMPLETE

- You get an answer.
- CoCo wraps your query results in a **CORTEX.COMPLETE** call.

```
"Based on this month's GitHub
```

```
data, the fastest-growing AI
```

```
project is X, driven by Y..."
```

- Natural language. Cited from data.



Step 5: Wire the Agent

You built this. CoCo was the co-pilot. You drove.

Your CoCo prompt

"Create a Cortex AI Agent called GitTrend. It should answer questions about trending GitHub repositories using the query we built. Give it a system prompt that defines its role and grounds it in the GH Archive data."

CoCo generates

A CREATE CORTEX SEARCH SERVICE on the repo descriptions

A Cortex Agent definition with tools pointing to your data

A system prompt that tells the agent what it knows and what it doesn't

Test queries to validate it's working

The result

A named agent > **GitTrend** — that any engineer on your team can query in natural language.

No SQL. No Marketplace subscription management. Just questions & answers.



RUN IT



Ask Your Agent

This same pattern works on your company's data — sales, support, product, ops.

Open CoWork. Find GitTrend.

Or test directly in CoCo. Either way > your agent is live!

Questions to try

"What's the fastest-growing AI project in the last 30 days?"

"What programming languages dominate trending AI repos?"

"Is there anything trending in agentic AI or MCP this month?"

"Compare the top 5 repos by stars — what do they have in common?"

What just happened

Your agent queried 4B GitHub events, aggregated real-time trends, & answered in plain English.

Zero SQL written by you. Grounded in real data. In under 60 minutes.



What You Built Tonight

The same stack deploys against any dataset in your organization.

1

AI Agent

- Built from scratch
- CoCo as co-pilot
- Yours to keep

107M+

GitHub Events

- Real data
- No ETL
- public S3, free

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SQL Written

- CoCo wrote every
- query and every
- Cortex wrapper

∞

Questions

- Ask anything
- In plain English
- CoWork or API



Let's **Stay Connected**

LINKEDIN



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MEDIUM



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RICHIE BACHALA · SR. MANAGER, SOLUTIONS ARCHITECTURE · SNOWFLAKE



Thank You

Workshop materials + checkpoints:

github.com/sfc-gh-rbachala/building-ai-agents-with-coco-workshop

Free Snowflake trial: <https://bit.ly/3SETPFF>

CoCo quickstart: docs.snowflake.com/coco

